STEAMROOT · USER REQUIREMENTS

SteamRoot user requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/steamroot		
Source	docs/03a-user-requirements.md @ 276b380		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First issue (kit 1.9.1 upgrade): 14 user requirements from the users and stakeholders in STR-PRB-001

SteamRoot user requirements

What the growers, operators and stakeholders of STR-PRB-001 need, written from their side. Product requirements that meet these needs are in STR-PRS-001 (`03-requirements.md`) and the links are in the generated traceability matrix STR-RTM-001 (`03b-traceability.md`). They are proposals to review with the first co-design partners (one market garden and one nursery near Irving, decided by Amish on 2026-10-02, not yet named). Method: STANDARDS section 22.

> **Safety:** SteamRoot is a fired steam generator. The safety needs UR-005, UR-007 and UR-011 take priority over the performance needs when they conflict.

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Performance	Clear weeds and disease to working depth	I need the weed seed bank and soil-borne disease cleared in the top 15 cm of a bed without herbicides or fumigants	High	Small organic farm or market garden; nursery (PRB Table 1)
UR-002	Performance	Treat beds at a useful rate	I need to steam beds fast enough that a working day treats a useful length of bed	High	Small organic farm or market garden (PRB Table 1)
UR-003	Performance	Steady steam from one fire	I need steam to keep coming from one fire without stopping to restart it	Medium	Operator (PRB Table 1)
UR-004	Cost	Run on wood I already have, sparingly	I need it to burn the prunings, hedgerow wood and pallets I already have, and not much of them, with no diesel or propane	High	Small organic farm (PRB, problem and constraints)
UR-005	Safety	Never a pressure vessel	I need a steam generator that cannot build pressure or burst, with little hot water stored in it	High	All users; boiler authority (PRB Table 2)
UR-006	Usability	Run for half a day between refills	I need to steam for a half-day session before refilling water	Medium	Operator (PRB Table 1)
UR-007	Safety	Protect me and the site	I need the fire, sparks, smoke and steam kept away from me, bystanders and dry vegetation	High	Operator; community garden (PRB Tables 1 and 2)
UR-008	Logistics	Tow it with what I have	I need to tow it with a small tractor, utility vehicle or car with a hitch, and move it between farms	High	Farm cooperative; community garden (PRB Table 1)
UR-009	Cost	Afford it, alone or shared	I need it to cost what a small farm or a cooperative can find	Medium	Small farm; farm cooperative (PRB Table 1)
UR-010	Usability	One trained operator, a helper nearby	I need one trained adult to run it, with a second person nearby while steaming	High	Operator (PRB constraints)
UR-011	Regulatory	Legal to build and run	I need the design to clear the boiler authority before anyone builds or fires it	High	Amish; Texas Department of Licensing and Regulation (PRB Table 2)
UR-012	Deployment and assembly	Buildable in a farm workshop	I need to build and repair it with stock steel, stock stainless tube and off-the-shelf safety parts	Medium	Open hardware community; farm workshop (PRB Table 1)
UR-013	Environment and durability	Fire outside, steam inside	I need to pipe steam into a polytunnel or greenhouse while the fire stays outside	Medium	Nursery or seedling grower (PRB Table 1)
UR-014	Maintenance			Low	

ID	Category	Title	Need	Priority	Source
		Open and documented	I need an open, documented reference design I can inspect, teach from and study		Extension educator or researcher (PRB Table 1)

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory. Priorities: High, Medium, Low.